# Zero-Knowledge Federated Learning with Lattice-Based Hybrid Encryption for Quantum-Resilient Medical AI

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***Abstract*—Hospital federated learning (FL) must resist gradient inversion, Byzantine poisoning, and Harvest-Now-Decrypt-Later (HNDL) risk under multi-year retention. ZKFL-PQ is a *compositional* stack: NIST ML-KEM-768 transport; dual-norm language $\mathcal{L}^{\mathrm{bind}}_{\tau_2,\tau_\infty}$ (Unruh $\ell_2$ NIZK + public $\ell_\infty$ clip on the same BFV plaintext); Unruh-lifted Enc-consistency; $(t,n)$-threshold BFV open with a PartialDecrypt FS accountability check; adaptive public $\tau_2$ and round-transcript binding; and default post-ZKP median. The artifact ships fused SEAL+threshold HE, shared-SIS Unruh, Keccak/SHA3 formal checks, and measured demos (with explicit library-default vs. latency $r$). Empirically: multi-seed baselines, UCI pipeline validation under the innovation stack, MedMNIST/`ConvNet28` smokes, backdoor ASR, and a sparse-poison dual-norm ablation ($100\%$ reject vs. $0\%$ for $\ell_2$-only). Complementary to Beskar-class secure aggregation—not a replacement.**

***Index Terms*—Post-quantum cryptography, federated learning, zero-knowledge proofs, homomorphic encryption, medical AI, Byzantine robustness**

## I. Introduction

Federated learning (FL) [1] keeps raw records at the edge, yet model updates remain vulnerable to gradient inversion [2], Byzantine poisoning [3], and HNDL attacks against classical TLS when medical archives outlive RSA/ECC [4]. Hospital consortia therefore need (i) post-quantum (PQ) session confidentiality, (ii) verifiable rejection of oversized updates, and (iii) aggregation that does not place a single curious server in sole possession of plaintext gradients.

Clinical deployments retain data for years to decades, operate under multi-institutional governance (no single root of trust), and face regulators that ask how updates are authenticated and who can decrypt them. Wrapping FedAvg in classical TLS is insufficient against adversaries that store ciphertexts today and open them later [4]. Cryptographic accept alone is also insufficient: updates whose $\ell_2$ norm respects a public threshold $\tau$ can still flip signs or implant triggers, so statistical robust aggregation must follow a successful proof.

*a) Design goals.:* We optimize for hospital consortia where (G1) ciphertext retention is measured in years, (G2) no single IT operator may hold a usable $sk$, (G3) oversized gradients must be rejectable without trusting the client, and (G4) in-bound Byzantine behavior that passes $\tau$ is handled by a robust aggregator rather than by tightening $\tau$ until honest sites fail. These goals motivate the four-layer stack rather than any single primitive.

*b) Contributions.:*

1) A layered protocol for $\mathcal{L}^{\mathrm{bind}}_{\tau_2,\tau_\infty}$ with Unruh Enc-consistency, PartialDecrypt-accountable threshold open, then post-ZKP median (Table I).
2) Dual-norm gating, adaptive public $\tau_2$, and transcript binding—with honest scope on which gadgets are Unruh vs. classical FS.
3) Concrete parameters (library defaults vs. latency demos), an *informal* composition theorem, Lean/Python/EasyCrypt checks, and measured payloads/latency.
4) A reproducible artifact [12] and empirical study: multi-seed baselines, innovation-pack UCI pipeline validation, MedMNIST/`ConvNet28` smokes, backdoor ASR, and dual-norm ablation (Table V).

## II. Threat Model and Claims

**Adversary:** (i) a network eavesdropper with future quantum computers against factoring/DLP; (ii) up to $f<N/2$ Byzantine clients that may submit arbitrary vectors and proofs; (iii) a curious aggregator that sees all protocol messages but does not hold $sk$ alone; (iv) up to $t-1$ colluding threshold decryptors, *plus* the possibility that one decryptor returns a malformed partial $\mu_i$.

**Trust.** Honest clients follow local SGD, $\ell_\infty$ clipping, and the Prove interface. Honest decryptors compute $\mu_i$ correctly (malicious partials are caught by PartialDecrypt NIZK and abort the open). The aggregator correctly runs Verify and the chosen robust aggregator.

**Claims:** ML-KEM-768 transport (FIPS 203 Level 3) [5]; dual-norm soundness for oversized / sparse witnesses under SIS + Unruh/RO; Unruh Enc-consistency so a proof cannot attach to a mismatched **ct**; threshold privacy *and* open correctness against a malicious partial; transcript non-swap across rounds.

**Scope.** We do not claim differential privacy or a drop-in replacement for Beskar-class full-stack secure aggregation [8]. ZKFL-PQ targets verifiable norm gating, multi-party decrypt, and robust post-filtering; pure $\ell_2$ alone does not stop sign-flip or in-bound backdoors (Table I).

TABLE I
THREAT COVERAGE BY LAYER (COMPOSITION IS THE CONTRIBUTION).

| Threat | ML-KEM | Dual-norm | Unruh-Enc | PD-NIZK | Median |
|---|---|---|---|---|---|
| HNDL / PQ transit | yes | – | – | – | – |
| Oversized update | – | $\ell_2$ | – | – | – |
| Sparse / $\ell_\infty$ spike | – | $\ell_\infty$ | – | – | help |
| Proof/ct split | – | bind | yes | – | – |
| Malicious partial $\mu_i$ | – | – | – | abort | – |
| Curious aggregator | – | – | – | thr. | – |
| Sign-flip / backdoor | – | no‡ | – | – | yes |

‡In-bound updates may still satisfy $(\tau_2, \tau_\infty)$; Table IV reports 0% crypto detection for sign-flip under $\ell_2$ alone.

TABLE II
POSITIONING VS. RELATED PQ / ROBUST FL LINES.

| System | PQ transport | Norm ZKP | HE / thr. dec. |
|---|---|---|---|
| Beskar [8] | yes (masking/DP) | – | secure agg. stack |
| Patel [9] | lattice HE | – | HE FL |
| Lu et al. [10] | hybrid PQ | – | HE FL |
| Krum [3] | statistical | – | – |
| zkML [11] | – | inference ZK | – |
| ZKFL-PQ (ours) | ML-KEM-768 | Unruh+bind | BFV$(t, n)$ |

## III. RELATED WORK

Beskar offers a mature PQ secure-aggregation stack with assisting nodes and DP [8]. Patel [9] and Lu et al. [10] study PQ-HE FL without ciphertext-bound norm ZKPs. zkML targets inference proofs rather than FL aggregation [11]. Classical robust aggregators (Multi-Krum, coordinate median, trimmed mean) remain effective against low-norm adversaries but do not provide PQ transport, ciphertext binding, or threshold open correctness. ZKFL-PQ therefore *composes* cryptographic accept with median/Krum rather than replacing either line: Hybrid+mean vs. Hybrid+median (Table IV) and dual-norm vs. $\ell_2$-only (Table V) make the division of labor measurable.

## IV. PROTOCOL DESIGN

### A. Round Structure

Each round $t$: the server broadcasts $w^{(t)}$, public $(\tau_2, \tau_\infty)$, and transcript hash $H_{t-1}$. Client $i$ runs local SGD, applies $\mathrm{Clip}_\infty$ then quantization into $\tilde{w}_i \in \mathcal{L}^{\mathrm{bind}}_{\tau_2,\tau_\infty}$, encrypts under BFV $pk$, produces Unruh Enc-consistency and Unruh norm proofs bound to $\mathbf{ct}_i$, and sends an ML-KEM-wrapped payload. The server verifies, threshold-opens with PartialDecrypt NIZK (abort on malice), applies coordinate-wise median, then publishes adapted $\tau_2$ and $H_t$ (Algorithm 1).

*a) Transport.:* ML-KEM-768 encapsulates a session key used to AES-CTR-wrap $(\mathbf{ct}_i, \pi_i)$. This protects in-transit confidentiality against classical and future quantum adversaries on the channel; it does not replace HE confidentiality against the aggregator, which follows from RLWE + threshold opening.

*b) Verification checklist.:* Verify returns 1 iff: (i) Unruh invertible-RO records are consistent for the norm proof; (ii) challenge bits match the recomputed digest of first messages, $\mathbf{ct}_i$, public $(\tau_2, \tau_\infty)$, and transcript hash; (iii) SIS algebraic checks hold against the shared $\mathbf{A}$; (iv) response $\ell_2$ norms respect $B$ and the proven/encrypted vector satisfies the public $\ell_\infty$ clip policy (same plaintext as Enc-consistency); (v) Unruh Enc-consistency accepts for every BFV chunk. Fail-closed: any failed check drops the client for that round.

### B. Quantization and Dual-Norm Gate

Quantize clips coordinates to $[-c, c]$, scales by $s$, and rounds into $\mathbb{Z}_t$ (artifact defaults $c$=1, $s$=$10^3$). Before prove+encrypt, honest clients apply a public $\ell_\infty$ clip $\tilde{w} \leftarrow \mathrm{Clip}_\infty(\Delta, \tau_\infty)$ and then prove $\|\tilde{w}\|_2 \le \tau_2$ on the *same* vector that is encrypted. Sparse spikes with $\|\cdot\|_2 \le \tau_2$ but $\|\cdot\|_\infty > \tau_\infty$ are rejected before opening; Enc-consistency prevents proving a clipped vector while encrypting an unclipped one. Adaptive public $\tau_2$ is updated from accepted-norm quantiles and bound into the next round's associated data together with a rolling transcript hash $H(\mathsf{prev})$.

### C. Bound Norm Language

**Definition 1** (Dual-norm bound language). *Let* $C = \mathsf{Commit}(\tilde{w}; \mathbf{r}) = \mathbf{A}(\tilde{w}\|\mathbf{r}) \bmod q'$, $\mathbf{ct} = \mathsf{Enc}_{pk}(\tilde{w}; \rho)$*, and public* $(\tau_2, \tau_\infty)$*. Then*

$$\begin{aligned}\mathcal{L}^{\mathrm{bind}}_{\tau_2,\tau_\infty} = \big\{(C, \mathbf{ct}, \tau_2, \tau_\infty) : \exists(\tilde{w}, \mathbf{r}, \rho)\\ C = \mathsf{Commit}(\tilde{w}; \mathbf{r}),\ \mathbf{ct} = \mathsf{Enc}(\tilde{w}; \rho),\\ \|\tilde{w}\|_2 \le \tau_2,\ \|\tilde{w}\|_\infty \le \tau_\infty\big\}. \quad (1)\end{aligned}$$

*Honest provers set* $\tilde{w} = \mathrm{Clip}_\infty(\mathrm{Quantize}(\Delta), \tau_\infty)$ *before Prove+Enc so both norms apply to the* same *ciphertext plaintext.*

*a) Σ-protocol [6].:* Commit masks $\mathbf{y}$ and message $\tilde{w}$; the Fiat–Shamir challenge is $e = H(C\|\mathbf{ct}\|T\|\tau)$; responses $(\mathbf{z}, \mathbf{r}_z)$ satisfy $T + eC \equiv \mathbf{A}(\mathbf{z}\|\mathbf{r}_z)$ and $\|\mathbf{z}\|_2 \le B$. Rejection sampling yields HVZK in the classical ROM. Acceptance uses only algebraic and norm checks—never a client-supplied Boolean.

*b) Unruh-oriented NIZK.:* Classical FS is not tightly QROM-secure [7]. We use parallel binary Unruh sessions (library default $r$=128; latency-sensitive demos often 4–64) with invertible-RO records $(\rho, H(\rho))$. Challenge bits are extracted from $H(\mathsf{UNRUH}\|\tau\|\mathbf{ct}\|\mathsf{first\text{-}msgs})$. All sessions share one SIS matrix $\mathbf{A}$ (not $r$ independent keys), which keeps prove/verify interactive at imaging-scale $d$ (one $256\times(d+128)$ matvec family instead of $r$ separate keys). Combinatorial uniqueness / $2^{-r}$ lemmas are machine-checked in Lean 4; executable game hops and a SHA3-instantiated EasyCrypt QROM library with bit-level Keccak-$f$[1600] lane lemmas (θ-column parity, ρ/π/χ/ι locality, packing invertibility, 24-round fold) accompany the artifact (`formal/run_formal_ci.py`).

*c) Enc-consistency (Unruh-lifted).:* Beyond hashing $\mathbf{ct}$ into the challenge, a dedicated Σ-gadget proves knowledge of coins $\rho = (u, e_0, e_1)$ such that the BFV encryption equations hold for the same plaintext chunk. We lift this gadget under the same Unruh transform as the norm proof

TABLE III
CRYPTOGRAPHIC PARAMETERS (RESEARCH STACK).

| Component | Setting | Notes |
|---|---|---|
| ML-KEM-768 | $n$=256, $k$=3, $q$=3329 | FIPS 203 L3 [5] |
| BFV degree | $n$=512 / 4096 | Lightweight vs. `classic128` |
| BFV moduli | $q \approx 2^{40}$, $t = 2^{12}$ | NumPy encoder; optional estimator / SEAL |
| Threshold | $(t, n)=(2, 3)$ partial decrypt | Server $sk=\perp$ |
| Unruh reps | $r$=128 default | Enc Unruh $r_{\text{enc}}$ parallel |
| Post-ZKP agg | median (default) | Multi-Krum / mean overrides |
| Dual-norm | $\tau_2 + \tau_\infty$ clip | Same vector for Prove+Enc |
| Open audit | PartialDecrypt NIZK | Abort on bad $\mu_i$ |
| Packing | full-vector chunks | All $d$ coordinates encrypted |
| SIS commit | shared $\mathbf{A}$ | One $256 \times (d+128)$ matrix / prover |

(parallel binary sessions + invertible RO records), closing the QROM-uniformity residual between the two ct-bound proofs. Enc-consistency first messages are absorbed into the norm proof's associated data.

### *D. BFV Aggregation and Threshold Opening*

Additive BFV yields $\mathsf{Dec}(\sum_i \mathsf{Enc}(m_i)) = \sum_i m_i$ under a noise budget growing roughly as $N \cdot O(\sigma\sqrt{n})$. Full-vector packing encrypts all $d$ coordinates as $\lceil d/n \rceil$ independent ciphertexts under the same $pk$; the ZKP covers the concatenated quantized vector, so partial packing cannot hide an unprotected suffix. After key generation, secret coefficients are Shamir-shared and the monolithic $sk$ is erased. Opening uses Lagrange-weighted partial decryptions with smudging: party $i$ returns $\mu_i = c_1 \star (\lambda_i \cdot s_i)$ (plus noise). Each partial is accompanied by a PartialDecrypt NIZK proving knowledge of $s_{\text{eff}}$ with $\mu_i = c_1 \star s_{\text{eff}}$; a tampered $\mu_i$ fails verify and aborts the open—extending threshold privacy to *open correctness*. Modular negacyclic multiplication with parallel per-chunk encryption keeps full-vector runs interactive on CPU for $d \approx 5 \times 10^4$. An optional TenSEAL/SEAL sidecar runs as `FusedSealThresholdHE`.

*a) Post-ZKP aggregation.:* Let $S$ be the set of clients that pass Verify. Default aggregation is the coordinate-wise median $\bar{\Delta}_j = \mathrm{med}\{\Delta_{i,j} : i \in S\}$. Multi-Krum and plaintext mean are available as overrides. Median is intentionally applied *after* cryptographic accept: it does not replace the ZKP, and the ZKP does not replace median.

**Theorem 1** (Informal composition)**.** *Under SIS binding of* $\mathsf{Commit}$*, RLWE security of BFV, and RO modeling of* $H$=$\mathsf{SHA3\text{-}256}$*: (i) honest members of* $\mathcal{L}^{\text{bind}}_{\tau_2,\tau_\infty}$ *are accepted except for rejection-sampling aborts; (ii) witnesses outside the dual-norm language are rejected except with Unruh/SIS advantage on the* $\ell_2$ *sessions (the* $\ell_\infty$ *gate is enforced by the public clip+Enc-consistency binding, not a separate Unruh range proof); (iii) Unruh Enc-consistency prevents proof/ct splits; (iv)* $t-1$ *decryptors learn nothing beyond t opens, while a malformed* $\mu_i$ *fails the PartialDecrypt FS check and aborts (ROM soundness for that gadget); (v)* $H_{t-1}$ *binds transcripts across rounds. Lean checks the combinatorial* $2^{-r}$ *Unruh counting lemma; EasyCrypt ships SHA3/Keccak theories with executable FIPS checks—not a fully discharged end-to-end EasyCrypt proof of the FL protocol.*

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**Algorithm 1** ZKFL-PQ round (innovation composition)

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**Require:** $w^{(t)}$, $(\tau_2, \tau_\infty)$, KEM keys, HE $pk$, decryptor shares, transcript $H_{t-1}$

Broadcast $w^{(t)}$, $(\tau_2, \tau_\infty)$, $H_{t-1}$
**for** each client $i$ **do**
  $\tilde{w} \leftarrow \mathrm{Clip}_\infty(\mathrm{Quantize}(\mathrm{LocalSGD}(w^{(t)}, D_i)), \tau_\infty)$
  $\mathbf{ct}_i \leftarrow \mathsf{Enc}_{pk}(\tilde{w}; \rho_i)$; $\pi_i^{\text{enc}} \leftarrow \mathrm{UnruhEnc}(\rho_i)$
  $\pi_i \leftarrow \mathsf{Prove}_{\mathrm{Unruh}}((\mathbf{ct}_i, \tau_2, \tau_\infty, H_{t-1}), \tilde{w})$
  send ML-KEM-wrapped $(\mathbf{ct}_i, \pi_i, \pi_i^{\text{enc}})$
**end for**
Keep $i$ iff dual-norm cryptographic verify returns 1
For each accepted $i$: threshold-open with PartialDecrypt NIZK (abort if fail)
$\bar{\Delta} \leftarrow \mathrm{Median}(\{\Delta_i\})$; publish $\tau_2 \leftarrow \mathrm{Adapt}(\{\|\Delta_i\|\})$, $H_t$
$w^{(t+1)} \leftarrow w^{(t)} + \bar{\Delta}$

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*b) Implementation invariants.:* The artifact [12] realizes Algorithm 1 with shared prove/encrypt dimension, crypto-only accept, full-vector packing, Unruh Enc-consistency, PartialDecrypt open, fused SEAL+threshold HE, default median, shared SIS $\mathbf{A}$, adaptive $\tau_2$, and transcript binding. **Parameter honesty:** library defaults target $r$=128 (norm) and HE preset `classic128_demo` ($n$=512, research-class moduli); latency demos and the innovation pack may use smaller $r/r_{\text{enc}}$ and must be read as such. Production-oriented $n$=4096 is selectable via `ZKFL_HE_PRESET=classic128` and is slower on NumPy.

## V. EVALUATION

### *A. Synthetic Multi-Baseline Study*

**Setup.** Synthetic 4-class task (1,000 samples, 784 features, Dirichlet $\alpha$=0.5), $N$=5, MLP with $d$=108,996, 10 rounds, seeds $\{42, \ldots, 46\}$. Attacks from round 4: (i) large-norm Gaussian; (ii) sign-flip scaled to $\|\cdot\|_2 = \tau$. Baselines: FedAvg, clip@$\tau$, Multi-Krum ($f$=1), HE-only, ZKP-only, Hybrid+mean (privacy path without robust agg), and Hybrid+median (default composition). Metrics: final accuracy and cryptographic detection rate (fraction of malicious updates rejected by Verify).

Large-norm poisoning collapses FedAvg/HE-only; clipping, Multi-Krum, and ZKP-style filters retain accuracy (Table IV, Fig. 1–2). Sign-flip within $\tau$ is accepted by pure $\ell_2$ and Hybrid+mean (0% crypto detection, Table I), while Multi-Krum and Hybrid+median recover accuracy by pairwise or coordinate-wise robust aggregation after cryptographic accept. The Hybrid+mean vs. Hybrid+median contrast is the empirical punchline for composition: privacy-preserving open without robust aggregation still fails on in-bound flips.

TABLE IV

MEAN±STD OVER 5 SEEDS (SYNTHETIC). ACCURACY (%) AND DETECTION.

| Method | Acc. (large) | Acc. (flip) | Det. (large) | Det. (flip) |
|---|---|---|---|---|
| FedAvg | 26.0±1.0 | 41.0±13.2 | 0% | 0% |
| Clip@$\tau$ | 100±0 | 40.3±12.7 | – | – |
| Multi-Krum | 100±0 | 100±0 | – | – |
| HE-only | 22.1±2.1 | 19.8±9.5 | 0% | 0% |
| ZKP-only | 84.9±30.2 | 40.9±13.2 | 97% | 0%‡ |
| Hybrid+mean | 87.0±26.0 | 40.1±12.6 | 97% | 0%‡ |
| Hybrid+median | 100±0 | 100±0 | 97% | 0%‡ |

‡Sign-flip respects $\tau$—$\ell_2$ proofs must accept (0% crypto detection). Hybrid+mean keeps plaintext mean after accept; Hybrid+median is the default composition and restores accuracy [12].

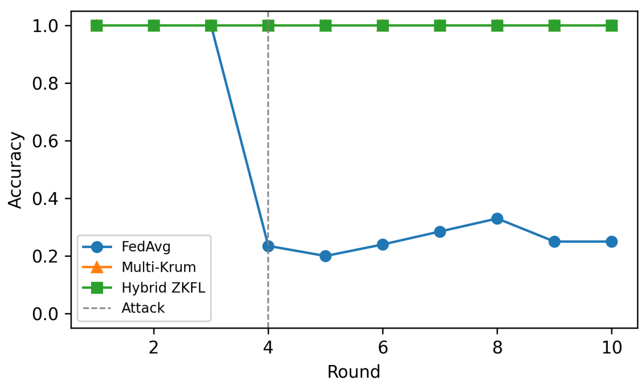


Fig. 1. Synthetic accuracy under large-norm activation (seed 42).

*a) Takeaway.:* Cryptographic detection is the right metric for oversized attacks; final accuracy after robust aggregation is the right metric for in-bound attacks. Reporting only one of the two understates the threat model in Table I.

### B. Scale, Medical, and Backdoor Studies

A compact-model scale study with $N$=20 clients and $T$=30 rounds reproduces the same qualitative pattern under both attacks: ZKP and hybrid reject oversized updates with high probability, while sign-flip still requires a robust aggregator. Wall-clock remains interactive on CPU for the compact MLP used in that study, which is the intended operating point for validating the crypto pipeline before scaling encoders.

On UCI Breast Cancer Wisconsin (569 samples, 30 clinical features; compact MLP with $d$=1554), we treat the task as a *pipeline validation* dataset—not a clinical efficacy claim. The innovation stack (norm Unruh $r$=16, Enc Unruh $r_{\text{enc}}$=4, PartialDecrypt NIZK, adaptive $\tau_2$, transcript, $\ell_\infty$ clip, threshold HE, median) yields 89.5% → 93.0% → 92.1% over three rounds, rejects the oversized client, and adapts $\tau_2 : 8 \to 4.3 \to 2.3 \to 2.0$ (`innovation_pack_results.json`; ≈20–23 s/round). Table V isolates the dual-norm gate on sparse poisons. PneumoniaMNIST/`ConvNet28` runs remain imaging smokes (≈93.3% at $d$≈51.6k) under reduced $r$ for CPU.

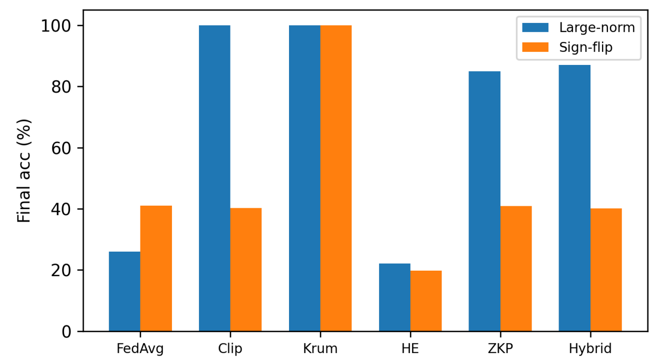


Fig. 2. Final accuracy: large-norm vs. sign-flip (5-seed means).

TABLE V

SPARSE-POISON ABLATION (SYNTHETIC): $\|\cdot\|_2 \le \tau_2$ BUT $\|\cdot\|_\infty > \tau_\infty$.

| Gate | Reject rate | Mean $\lvert\text{agg}_0\rvert$ |
|---|---|---|
| $\ell_2$-only | 0% | 0.899 |
| Dual-norm $(\tau_2, \tau_\infty)$ | 100% | 0.003 |

≈265× spike reduction on the poisoned coordinate (`innovation_pack_results.json`).

Trigger backdoors that remain inside $(\tau_2, \tau_\infty)$ are only partially mitigated by median (ASR ≈55% vs. ≈98% for FedAvg/$\ell_2$; Fig. 3). Dual-norm removes sparse spikes *before* opening; residual ASR shows that diffuse in-bound poisons still need stronger robust aggregators or DP—consistent with complementary—not replacement—positioning vs. Beskar.

### C. Communication Cost

Full-vector HE dominates bandwidth: $O(N\lceil d/n\rceil n \log q)$ plus Unruh $O(r \cdot (d + \ell))$ and Enc-Unruh $O(r_{\text{enc}}\lceil d/n\rceil)$. Table VI separates payload from wall-clock so demo $r$ is not confused with the $r$=128 library default.

## VI. SECURITY DISCUSSION

Honest members of $\mathcal{L}^{\text{bind}}_{\tau_2,\tau_\infty}$ pass Unruh algebraic/norm checks except for rejection-sampling aborts. Oversized $\ell_2$ witnesses fail sessions with challenge bit 1 except with small Unruh/SIS advantage; library default $r$=128 targets a combinatorial ≈ 128-bit Unruh class, while demos may use smaller $r$ at linear proof-size cost. The $\ell_\infty$ policy is cryptographic only insofar as Enc-consistency binds the clipped plaintext to $\mathbf{ct}$—it is not advertised as a standalone Unruh range proof. Unruh Enc-consistency aligns the Enc gadget with the norm proof's QROM story; PartialDecrypt NIZK remains a classical-FS accountability check (explicit ROM assumption). BFV+Shamir hide updates from the aggregator and $t-1$ parties. We do not claim DP.

*a) Attack games (informal).: Oversized/sparse:* Verify = 1 outside $\mathcal{L}^{\text{bind}}_{\tau_2,\tau_\infty}$. *Split:* proof verifies against $\mathbf{ct}' \neq \mathbf{ct}$. *Malicious partial:* forged $\mu_i$ accepted. *Curious open:*

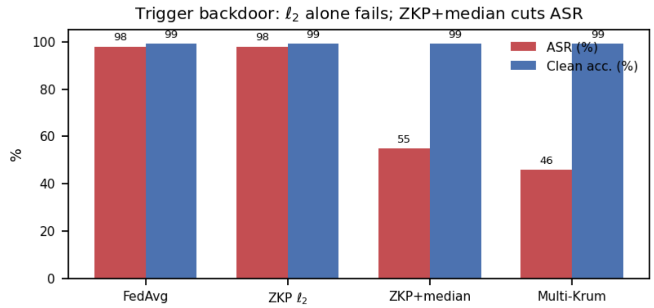


Fig. 3. Backdoor ASR vs. clean accuracy under FedAvg, $\ell_2$-ZKP, ZKP+median, and Multi-Krum.

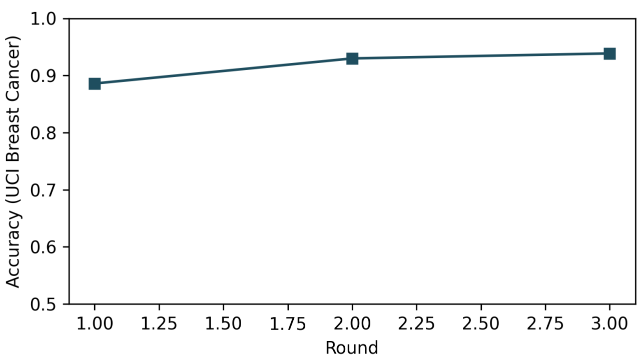


Fig. 4. UCI Breast Cancer accuracy under the full target stack.

aggregator+$<t$ decryptors recover plaintext. Theorem 1 states the informal composition under the listed assumptions.

*b) Layering and deployment.:* Table I is the punchline: composition, not a single primitive. Hospital deployment: site clients; coordinator verifies; $t$-of-$n$ opens with PartialDecrypt abort; median updates the model; adaptive $\tau_2$ and $H_t$ publish for the next round. UCI innovation-pack wall-clock (≈20 s/round) is acceptable for daily/weekly training; `ConvNet28` remains SGD-dominated once Unruh/HE use shared SIS and modular BFV.

## VII. Limitations and Reproducibility

EasyCrypt Keccak lane lemmas are axiomatized with a FIPS executable checker—not a full algebraic discharge without Python. PartialDecrypt NIZK is ROM-FS (not yet Unruh-lifted). Demo HE $n{=}512$ is a research preset, not a HomomorphicEncryption.org Classic-128 certificate. Innovation-pack $r{=}16/r_{\text{enc}}{=}4$ are latency settings; cite library defaults when claiming 128-bit Unruh class. Residual backdoor ASR under median (≈55%) shows dual-norm+median are insufficient alone against diffuse triggers. UCI/MedMNIST results validate the crypto pipeline, not bedside performance. Reproduce: `run_innovation_pack.py`, `run_target_protocol.py`, `formal/run_formal_ci.py`.

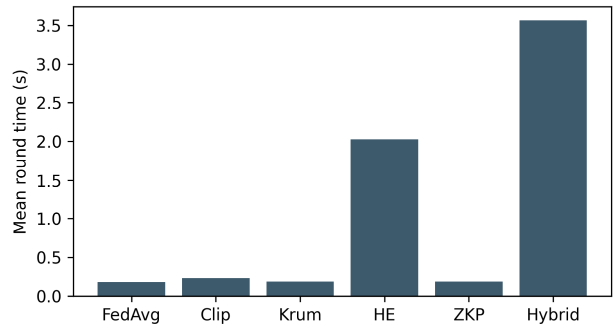


Fig. 5. Mean round time (synthetic hybrid ∼20× FedAvg).

TABLE VI
Payload / latency notes (artifact; demo $r$ may be $<$ library default).

| Setting | Msg or time | Notes |
|---|---|---|
| Synthetic hybrid | ∼100 KB | Reference accounting |
| UCI target ($r{=}64$, $n{=}512$) | ≈4.7–5.9 MB | Legacy target JSON |
| UCI innovation pack | ≈20–23 s/round | Denser Unruh-Enc+PD-NIZK |
| `ConvNet28` smoke ($r{=}4$) | ≈2.4 MB | $d{\approx}51.6$k |
| HE $n{=}4096$ preset | ≈2× ct vs $n{=}512$ | Slower on NumPy |

## VIII. Conclusion

ZKFL-PQ is a *compositional* PQ medical FL stack: dual-norm language $\mathcal{L}^{\text{bind}}_{\tau_2,\tau_\infty}$, Unruh Enc-consistency, PartialDecrypt-accountable threshold open, adaptive $\tau_2$, transcript binding, and post-ZKP median. Empirically, sparse $\ell_\infty$ spikes are rejected (100% vs. 0% under $\ell_2$-only), oversized clients are caught, and UCI pipeline accuracy stays competitive under denser proofs. The contribution is explicit threat-layering with an artifact—complementary to Beskar-class secure aggregation. Artifact: https://github.com/edlansiaux/pq-zkfl-medical.

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